

Generation Interconnection System Impact Study Report

For Queue Project Q16-0682

North Bristol 138 kV

November 26, 2017

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q16-0682. The Interconnection Customer proposes a 35.7 MW battery energy storage facility interconnecting at the North Bristol 138 kV substation in the Tennessee area. The request was received on September 28, 2016. The study evaluated system intact and single contingency conditions using the regional planning case, including 313 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 8 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$897,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
DILLARD-WILTON - ROGER SUBSTATION 138 KV Line	NORTH BRISTOL - FORNEY SW SUBSTATION 138 KV	120.7	183	17.3
FORNEY SW - N BRISTOL 138 KV Line	MICHIGAMME FALLS HY - NORTH BRISTOL 138 KV	116.7	252	33.2
MICHIGAMME FALLS HY - ROGER 138 KV	FORNEY W - N BRISTOL 138 KV	109.3	215	17.3
N LAYTOR - RADFORD ARMY AMMUNITION PLANT 138 KV Line	HINTON AP - LOWMOOR 345 KV	106.4	172	5.5
POMONA HTS - LYTLE 138 KV line	LOMOOR - HINTON AP 345 KV	106.0	221	5.5
SYSQUOC - LAKE FORREST 138 KV L	LOWMOOR - HINTON AP SUBSTATION 345 KV	105.5	200	5.5
LAKE FORREST - LYTLE 138 KV	LOWMOOR - HINTON AP 345 KV	104.6	213	5.5
MICHIGAMME FALLS HY - N BRISTOL 138 KV	FORNEY SW - NORTH BRISTOL 138 KV	100.8	301	35.9

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
ROGER SUBSTATION - DILLARD-WILTON 138 KV	Rebuild line with higher-capacity conductor	\$4,557,000	7% (\$334,000)
FORNEY SW - NORTH BRISTOL 138 KV	Rebuild line with higher-capacity conductor	\$3,366,000	7% (\$249,000)
MICHIGAMME FALLS HY - ROGER 138 KV	Reconductor line	\$2,361,000	7% (\$173,000)
N LAYTOR - RADFORD ARMY AMMUNITION PLANT 138 KV Line	Reconductor line	\$644,000	2% (\$10,000)
LYTLE - POMONA HEIGHTS 138 KV line	Reconductor line	\$1,239,000	2% (\$19,000)
LAKE FORREST - SYSQUOC 138 KV	Reconductor line	\$1,804,000	2% (\$28,000)
LAKE FORREST - LYTLE 138 KV	Reconductor line	\$1,413,000	2% (\$22,000)
N BRISTOL - MICHIGAMME FALLS HY 138 KV	Reconductor line	\$833,000	7% (\$62,000)
	Total allocated to project		\$897,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.